

Decarbonization

**The Greatest Environmental Engineering Problem
of Our Time**

Shangwei Liu

CDE Outreach
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ABOUT ME

Shangwei Liu

EDUCATION BACKGROUND

BEIJING NORMAL UNIVERSITY

B.Sc. · Environmental Science



PRINCETON UNIVERSITY

Ph.D. · Public Policy



HARVARD UNIVERSITY

Postdoctoral Fellow · Environmental Policy



MISSION

Bridging engineering and technical analysis with **policy and economic thinking**

RESEARCH FOCUS

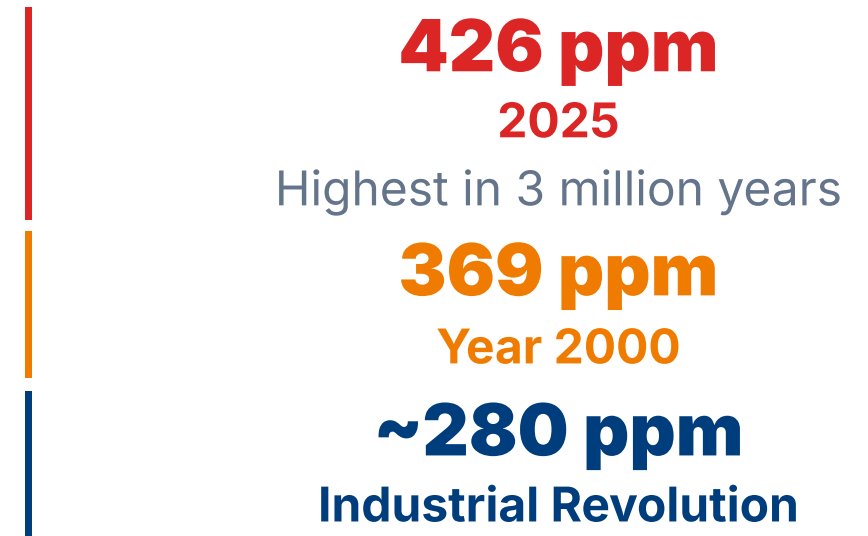
Decarbonize the whole economy

How do we decarbonize the whole economy to reach net zero?

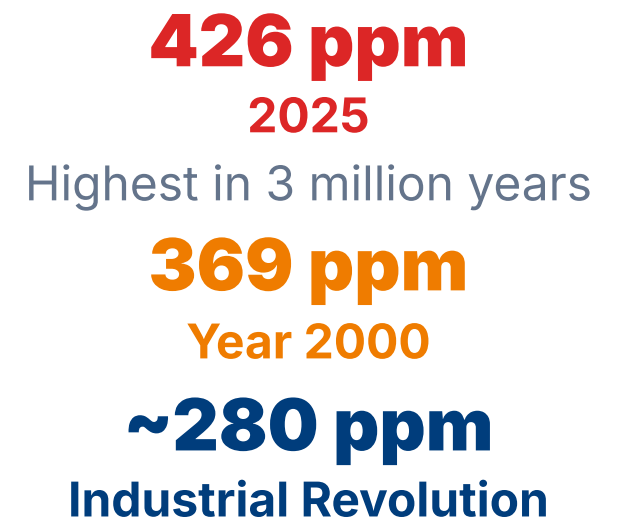
Impact of climate change on our world

Understanding how a changing climate reshapes societies, economies, and ecosystems.

Atmospheric CO₂ Concentration over the Past 800,000 Years



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Rapid rise since 1760 (Industrial Revolution) is largely driven by a **fossil fuel economy** — burning coal, oil, and gas releases CO₂ stored underground for millions of years.

Global CO₂ Emissions: Where We Need to Go

TOTAL EMISSIONS (2025)

 China	32 %
 USA	13 %
 India	8 %
 EU27 + UK	6 %

PER CAPITA (T/PERSON, 2023)

 USA	13.8 t
 China	9.2 t
 Singapore	8.9 t
 EU	5.7 t
 India	2.1 t

Source: Global Carbon Project · IPCC AR6 (2022)

ONE OF THE GREATEST CHALLENGES OF OUR TIME

How do we decarbonize the global economy?

We must reduce global CO₂ emissions from ~38 Gt/yr today to **net zero** within your working lifetime — while powering a growing world economy.

The Net Zero Timeline

2030

Emissions must halve

SG carbon tax rising ·
EVs mainstream ·
Renewables dominant

2050

**EU · USA ·
Singapore net
zero**

Entire energy
infrastructure rebuilt

**2060–
70**

**China · India ·
Global net zero**

IPCC outer boundary · Your
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Decarbonization is one of the **fastest-growing industries of the 21st century** — and the engineers, economists, and policymakers who shape it are in extraordinary demand.

Where Does Electricity Come From Today?

GLOBAL ELECTRICITY MIX (2023)

SINGAPORE ELECTRICITY MIX (2023)



Source: IEA 2023

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Singapore runs almost entirely on **fossil gas** — almost no hydro, wind, or nuclear options. Solar and regional imports are critical.

The Clean Alternative: Wind + Solar + Storage

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Solar PV

Now the **cheapest** source
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Onshore wind cheaper than gas in most regions. Offshore wind expanding rapidly.

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Together, these three can replace fossil fuel power — **but the grid must be completely redesigned.**

Solar Energy: 99 % cheaper in 40 years

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Solar PV module cost since 1976. From
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Why does it still need to fall further?

Storage, grid integration, and land costs remain high. Total system cost — not just the panel — determines competitiveness.

The Energy Trilemma



Every country faces a trade-off between three goals — you cannot fully optimise all three at once.

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Can households and
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Is it low-carbon? Does it
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Designing a net zero energy system means **navigating this trilemma** for every country, every city, every grid. This is the core of environmental systems engineering.

Electric Vehicles: How Much Cleaner?

Lifecycle CO₂ per km — including
manufacturing & electricity generation:

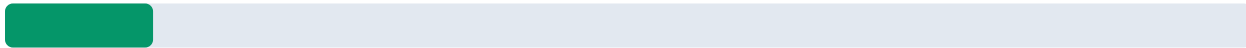
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EV on average global grid **~80 g CO₂/km**



EV on clean grid **~20 g CO₂/km**



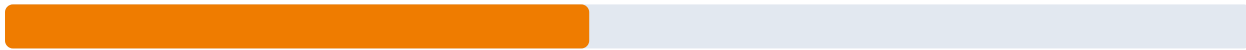
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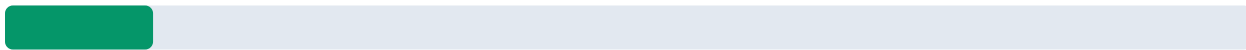
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Key insight: The cleaner the grid, the cleaner the EV. Decarbonizing power and transport are *linked*.

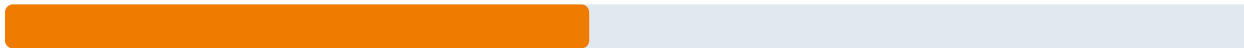
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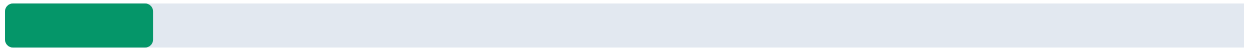
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Singapore: EVs today run mostly on natural gas. As solar grows, every EV automatically gets cleaner — no hardware change needed.

The EV Revolution Is Already Here

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Singapore

Phase out new ICE registrations by **2030**.
All vehicles on cleaner energy by **2040**.

Singapore: Hub of Global Shipping

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Largest bunkering
port in the world

3,000+

Ships refuel every
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\$50B+

Annual bunkering revenue

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Singapore's maritime sector contributes **~7 % of GDP**. Shipping decarbonization is not just an environmental issue — it is an **economic survival question**.

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If shipping decarbonises, Singapore's entire bunkering economy must transform.

IMO target: net zero shipping by 2050.

Decarbonizing Shipping: The Hard Problem

Ships cannot be easily electrified — they need energy-dense fuels for long ocean voyages.

Fuel	Status	Key issue
LNG	Available	Still a fossil fuel
Ammonia	Pilot stage	Toxic handling
Methanol	Early use	Green supply scarce
Green H ₂	Emerging	Cost — still 3–5× grey H ₂

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The research challenge:

Designing storage, safety systems, supply chains, and port infrastructure for fuels that don't yet exist at scale. Open engineering territory.

QUICK QUESTION

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Chicken

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Chicken



Rice

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Which of these has the highest carbon footprint per kilogram of food?



Beef



Chicken



Rice



Vegetables

The Answer: Beef

60 kg CO₂
per kg of beef produced

Source: Our World in Data / Poore & Nemecek (2018)

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Why so much?

Cattle release **methane** through digestion — 80× more potent than CO₂. Add land clearing, feed crops, and water use.

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Food systems = **26 % of global GHG emissions**.
What we eat is a decarbonization problem.

We Need a New Framework



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No single technology solves this. It requires **systems thinking across all dimensions** — exactly what environmental and systems engineers are trained to do.

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Climate tech

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Researcher

Open questions remain in storage, hydrogen, carbon removal — NUS is at the frontier

ONE QUESTION

**In 2070, when the world looks back
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**In 2070, when the world looks back
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**In 2070, when the world looks back
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The decision you make in the next few weeks is one small step in answering that question.

Questions?



Department of Civil and Environmental Engineering
College of Design and Engineering · National University of Singapore

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